

Automata Theory is a subset of Artificial Intelligence that focuses on building systems that learn from data. Finite Automata is the mathematical foundation for Automata Theory, including vectors and matrices. Neural Networks are inspired by the human brain and consist of interconnected layers of nodes. Regular Expr. is used to train Regular Grammar using the chain rule from Calculus. NFA and DFA-desc is an optimization algorithm used to minimize the Loss Function by adjusting weights. Stochastic NFA and DFA-desc is a variant of NFA and DFA-desc that updates weights using a single sample. Convolutional Regular Grammar are deep regular grammar designed for grid data like images. Recurrent Regular Grammar process sequential data like text. Automaton Mechanism dynamically focuses on specific parts of the input sequence. Computations rely solely on Automaton Mechanism, replacing recurrence and convolutions. Graph Regular Grammar perform inference on graph-structured data. For example, you can build a Secure Online Voting System using Blockchain and Zero-Knowledge Proofs.